

Technical Report: The Daily Reflection Knowledge Engine

Subject: Knowledge Engineering for Employee Mindset Optimization

Project: Deterministic Decision Tree for Workplace Reflection

1. Executive Summary

The Daily Reflection Tree is a deterministic expert system designed to facilitate high-quality cognitive reframing for employees at the end of the workday. Unlike probabilistic Machine Learning models, this tool utilizes Knowledge Engineering to ensure a consistent, auditable, and psychologically grounded experience. By navigating a 25-node state machine, users are guided through three critical psychological axes: Locus of Control, Growth Mindset, and Radius of Impact.

2. Psychological Foundations (The Three Axes)

The tool's logic is built upon three proven behavioral science frameworks:

Axis 1: Locus of Control (Rotter, 1966)

The first phase of the tree identifies the user's "Locus of Control."

External Locus (Victim): Users who attribute their day's outcomes to "bad tools" or "unclear instructions."

Internal Locus (Victor): Users who focus on their own reactions and time management.

Engineering Goal: Nodes 2 through 7 use "Reframing Nodes" to gently nudge users from an external to an internal locus, promoting personal agency.

Axis 2: Orientation and Mindset (Dweck, 2006)

The second phase assesses "Contribution vs. Entitlement" using the Growth Mindset framework.

Fixed Mindset: Approaching tasks as routine "box-checking."

Growth Mindset: Treating challenges as opportunities for mastery.

Engineering Goal: Nodes 9 through 14 identify "knowledge gaps" and reframe them as "technical debt" or "learning opportunities."

Axis 3: Radius of Impact (Maslow / Altrocentrism)

The final phase measures the "Radius of Impact"—how far the user's focus extends beyond themselves.

Self-Centric: Focusing solely on one's own ticket or task.

Altrocentric: Ensuring work is documented for the "next person" and supporting teammates.

Engineering Goal: Nodes 16 through 20 evaluate if the user is operating as a "Solo Specialist" or a "Systemic Contributor."

3. Engineering Architecture

Deterministic State Machine

The tool is implemented as a Deterministic Decision Tree. This was chosen over Generative AI for several key reasons:

Consistency: Every user who provides the same inputs will receive the same high-quality reflection.

Zero-Cost Runtime: The tool requires no expensive API calls or GPU inferencing; it runs locally as a lightweight JSON-parsed script.

Auditability: Because the logic is stored in a static tree.json file, the decision-making process is fully transparent and can be reviewed by human experts to ensure psychological safety.

Data Structure and Signals

The tree utilizes a Signal-Based Tally System. As users navigate the nodes, invisible "signals" (e.g., int_locus, growth, altro) are collected. The final decision node (Node 21) acts as a logic gate, using these tallies to select one of three final summaries:

The Victor: High internal locus and high growth mindset.

The Solo Specialist: High growth but narrow radius of impact.

The Passenger: High external locus and low engagement.

4. Conclusion

The Daily Reflection Tree demonstrates that sophisticated behavioral change does not always require complex AI. By encoding human expertise into a structured JSON format, we have created a scalable, private, and effective tool for professional development. This "Knowledge Engineering" approach ensures that the "wisdom" of the tool remains stable, predictable, and aligned with organizational values.

Would you like me to expand on any specific section, such as the "Auditability" part or the "Reframing Nodes" logic?